Estimating hidden post views from reactions: transferring a model across platforms

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draft, November 9, 2026

Abstract

On Facebook and Instagram, media-monitoring data contain no view counts. We test whether views can be estimated from likes and comments by a model that has never seen a view count from these platforms. A linear regression on likes and comments, trained only on X and YouTube, estimates the views of 402 Facebook videos with a typical error of $\times 2.43$ (MAE 0.385 on the $\log_{10}$ scale), the same as the same model trained on Facebook, and of 642 Instagram reels with $\times 2.72$, worse than the same model trained on Instagram ($\times 2.41$). The median views of the source platforms are off by $\times 9$–$\times 20$ there. In practice the model transfers the ratio of views to likes: it beats a fixed likes multiplier on three of six target platforms, is not worse on two and is worse on X. The transfer needs reactions read at the same moment as the views; on Stack Exchange only the ordering transfers, and the level is worse than a constant median. The results rest on 20 and 11 brands respectively, and source and target views were read at different post ages (23 against over 300 days), which may partly mask differences.

1 Introduction

Media monitoring companies such as Brand24 report to their clients how many people may have seen a mention of their brand. The simplest reach measure, the view count, is often not visible: platforms limit what they show publicly, and the scope of this data differs between platforms and post formats and changes over time. Reach is then approximated by a rule based on followers or reactions, which says nothing about how wrong it may be.

Predicting content popularity is widely studied [1], also across services: Shulman et al. [2] transfer a model between music, book, photo and link datasets, and Roy et al. [3] transfer knowledge from Twitter to YouTube videos. These works, however, predict future popularity and have popularity data from the target platform on which the model is trained or calibrated. Our task is transfer without labels from the target domain [4]: we estimate current views from visible reactions on a platform from which the model gets not a single view count in training. We collect views from that platform separately and use them only for evaluation. We found no work that tests transferring a view model across platforms in this way.

Our contributions:

- we show that a plain linear regression on likes and comments, trained only on X and YouTube, is enough: it estimates the views of Facebook videos with an error of $\times 2.43$, the same as a model trained on Facebook, and of Instagram reels with $\times 2.72$ (model trained on Instagram: $\times 2.41$);
- we compare it fairly with simple rules on six target platforms and show when the transfer works (reactions read together with views) and when it fails (X, the level on Stack Exchange);
- we give an uncertainty interval for every post, well calibrated on the same platform; out of platform it needs calibration on a few brands from the target platform;
- we show applications: a breakout alert, a brand ranking and media value.

2 Data

We use 12 500 mentions of 348 brands from five platforms, provided by Brand24 (2 500 per platform; a brand may appear on several platforms; 85% of mentions were published between April and August 2026, and 85% are in English). For TikTok, YouTube and X the views are in the export. The export also has a reach column, which Brand24 computes by a rule from reactions and followers (according to our reconstruction) and which replaces views on Facebook and Instagram. For Facebook and Instagram we collected the views ourselves in November 2026, together with likes and comments from the same moment. Of 470 Facebook videos, 68 had been removed, leaving 402 from 20 brands. Of 976 Instagram reels, 261 had been removed, 19 had no view counter and 54 had hidden likes, leaving 642 from 11 brands. Ordinary posts on these platforms have no

view counter, so the results apply only to videos and reels. We also re-read the views and reactions of 2 013 X posts, 1 376 YouTube posts and 426 TikTok posts (Section 5).

Table 1: Data by platform.

Platform	Brands	Authors	Source of views	Posts w/ views	Median views	Views/like
TikTok	266	2 411	Brand24 export	2 491	729	25
YouTube	273	2 041	Brand24 export	2 441	433	26
X	252	1 658	Brand24 export	1 803	27	22
Facebook	20	≥182[a]	own read (videos)	402	1 413	35
Instagram	11	?[b]	own read (reels)	642	504	53

a: author known for 74% of videos. b: author known for only 11% of reels (59 authors among 72). Views/like: median ratio over posts with at least one like.

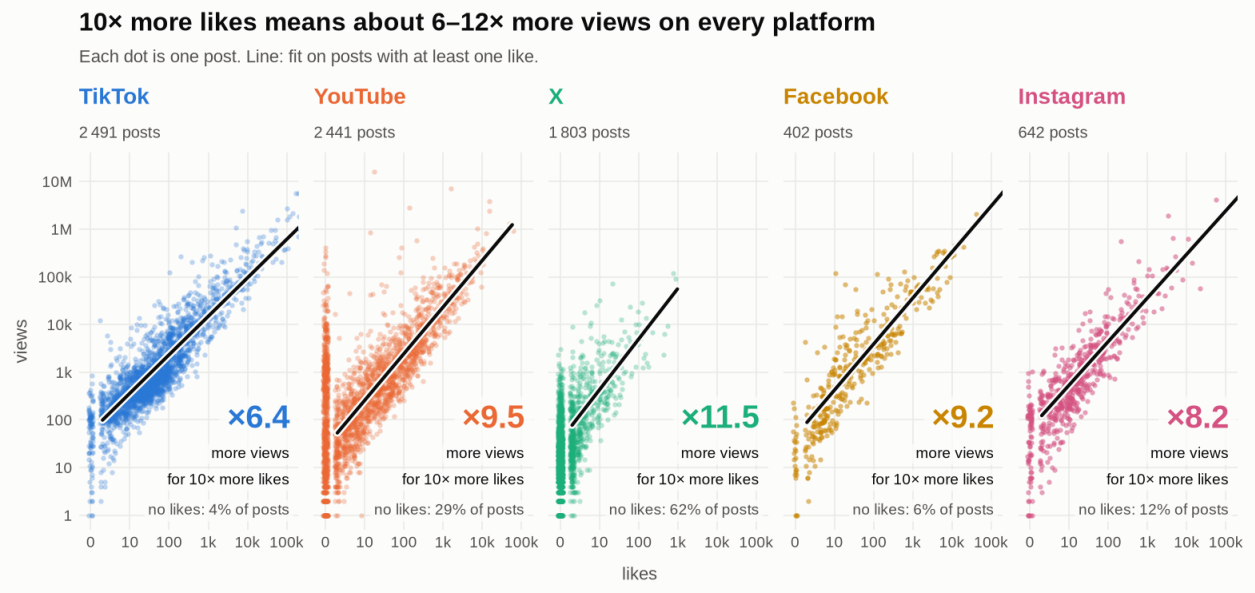


Figure 1: Views versus likes, separately for each platform.
Both axes on a log scale; posts without likes at zero. Line: fit on posts with at least one like; number: how many times more views a post with 10 times more likes has (6–12). TikTok, YouTube, X: Brand24 export; Facebook (videos), Instagram (reels): our read.

3 Method

We predict $y = \log_{10}(\text{views})$. We report the error as MAE on this scale and as the *typical error*, the factor 10^{MAE} (MAE 0.30 ≈ ×2; the geometric mean ratio of estimate to truth). Agreement in ordering is measured by Spearman's rank correlation between estimated and true views. Every model is compared with two rules: the median views and a *fixed likes multiplier* (views = k·likes, k the median ratio of views to likes in the training data, about 25). Confidence intervals come from a bootstrap over brands, and differences between models on the same posts from a paired bootstrap over brands.

Validation. We join posts into groups if they share a brand, an author or an identical text (an author sometimes writes about several brands, and the same text is sometimes pasted on different accounts), and split whole groups into five folds. Every evaluation is thus on a brand and an author the model has not seen. Method comparisons (Table 4 and Fig. 4B) use a simpler split by brand alone; their numbers differ from Fig. 2 by at most 0.012 (CatBoost on X: ×2.95 against ×3.03), and the ordering of methods does not change.

Two scenarios, marked in every figure. *Same platform*: CatBoost trained on posts from the given platform (TikTok, YouTube, X), evaluated on brands and authors it has not seen. *Out of platform*: ridge regression that sees not a single post from the target platform in training (Facebook, Instagram, Stack Exchange; in Tables 2 and 3 also TikTok, YouTube and X in turn); target views are used only for evaluation.

Models. On the same platform we use CatBoost [5] on followers, reactions and missingness flags (a separate feature says whether a field was empty, because missing is not zero). Out of platform we use ridge

regression [6] on log(1 + likes) and log(1 + comments), the two fields every platform has, trained on X and YouTube together. With the target platform left out it transfers better than CatBoost and MLPs (by 0.014–0.062 MAE on each of the five platforms, also after data augmentation). The main Facebook and Instagram numbers use the Brand24 export for X and YouTube; the variant with a re-read source (Section 5) is reported separately, because it was tested after part of the reels had been seen. Per-post 50% and 90% intervals come from conformalized quantile regression (CQR) [7]: a quantile model gives a preliminary interval, and its width is corrected on a held-out part of the data so that coverage matches the target.

4 Results

Main results: how far off we are on a single post

		Posts / brands	**Our error**	Within ×2	Constant median
Same platform	TikTok	2,491 / 266	×**1.87**	67%	×4.6
	YouTube	2,441 / 273	×**2.90**	43%	×8.8
	X	1,803 / 252	×**3.03**	40%	×4.6
Out of platform (0 target posts)	Facebook	402 / 20	×**2.43**	53%	×19.9
	Instagram	642 / 11	×**2.72**	40%	×8.7

Our error: typical error 10^{MAE}, e.g. ×2.43 means the estimate is usually 2.43 times too high or too low. Constant median: one number for every post (the median of the same platform, and out of platform the median of X and YouTube).

Same platform. On TikTok, YouTube and X the typical error is ×1.87, ×2.90 and ×3.03 (MAE 0.273 / 0.462 / 0.482; Fig. 2). A fixed likes multiplier gives ×2.23, ×4.06 and ×3.63 here, so CatBoost beats it on every platform by 0.075–0.15 MAE (the confidence intervals of the differences exclude zero).

Out of platform. On Facebook and Instagram the model has not seen a single post from these platforms, and its error is MAE 0.385 (×2.43, 95% CI ×2.05–×3.03) and 0.435 (×2.72, ×2.49–×3.05). On Facebook the error is the same as when the same model is trained on Facebook (×2.42; difference 0.001, 95% CI [−0.03, 0.03], i.e. at most ×1.07 either way). On Instagram it is clearly worse than a ridge trained on Instagram itself (×2.41; difference 0.054, 95% CI [0.021, 0.093]), and 72% of reels are underestimated, because the median reel has 53 views per like, against about 25 on X and YouTube (Table 1). The result does not depend on choosing X and YouTube: the median over all 15 training sets without the target platform is ×2.58 on Facebook and ×2.65 on Instagram. The ordering of posts is preserved well (Spearman 0.91 and 0.86, within a single brand 0.80 and 0.83), but likes alone order them equally well. A typical brand has a larger error than the whole: the median over brands is ×2.7 on Facebook and ×2.9 on Instagram. Source and target also differ in read age (Section 8), which may partly mask differences between platforms.

Regression versus a fixed multiplier. In practice the regression transfers the ratio of views to likes, so it must be compared with the multiplier alone, fitted on the same data (Table 2). The regression is better on TikTok, Instagram and Stack Exchange, not better on Facebook and YouTube (the confidence intervals reach zero), and worse on X, where 62% of posts have no like and comments pull the estimate the wrong way. Both are 2–8 times more accurate than the median of the source platforms on every video platform, but on Stack Exchange the median beats both.

Table 2: Out of platform: the regression versus simple rules fitted on the same source platforms.
Typical error (×); MAE difference regression − multiplier with a paired brand-bootstrap 95% CI (negative: regression better). Source: X + YouTube; for X and YouTube as targets, the other four platforms. Median: one number for all posts, the median views of the source.

Target	Source	Regression	Likes multiplier	Difference [95% CI]	Source median
TikTok	X + YT	×2.05	×2.22	−0.034 [−0.038, −0.029]	×11.0
YouTube	other 4	×4.08	×4.03	+0.004 [0.000, 0.009]	×8.9
X	other 4	×4.37	×3.45	+0.103 [0.084, 0.120]	×22.1
Facebook	X + YT	×2.43	×2.42	+0.002 [−0.017, 0.022]	×19.9
Instagram	X + YT	×2.72	×2.94	−0.034 [−0.047, −0.020]	×8.7
Stack Exchange	X + YT	×3.14	×5.33	−0.229 [−0.240, −0.207]	×2.1

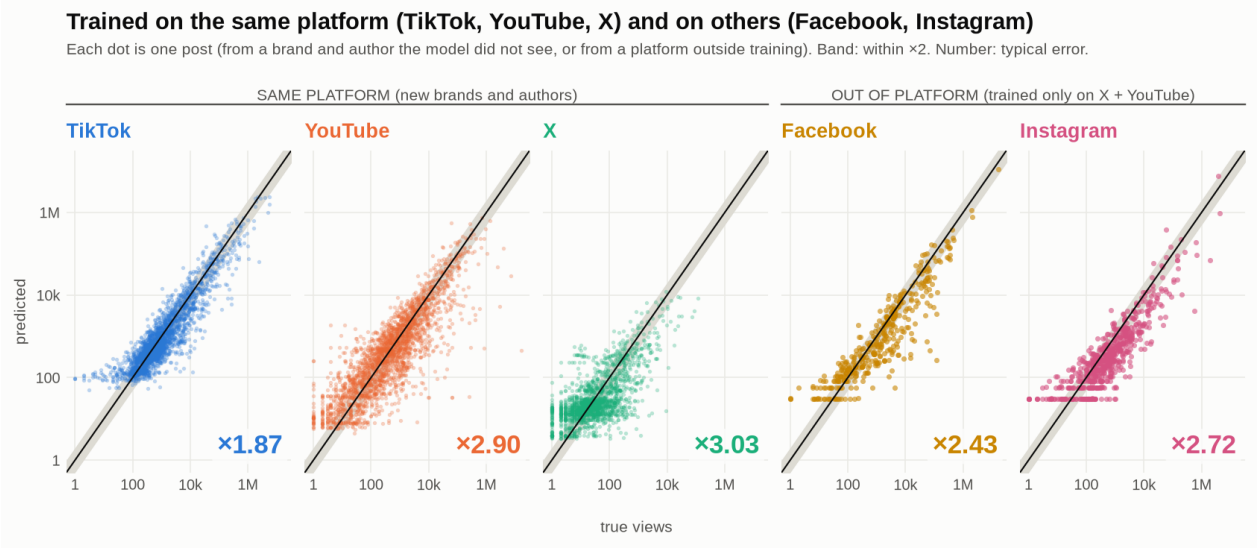


Figure 2: Estimated versus true views.
Each dot is one post. TikTok, YouTube, X: CatBoost on followers, reactions and missingness flags, trained on the same platform, validation with no shared brands, authors or texts; truth from the Brand24 export. Facebook (402 videos, 20 brands), Instagram (642 reels, 11 brands): ridge on log likes and comments trained only on X and YouTube, without any post from the target platform; truth from our read. Grey band: estimate within ×2. Number: typical error 10^{MAE}.

Who is estimated badly. On every platform, also out of platform, the model overestimates posts with few views (about ×2 below 100 views) and underestimates large ones. This is regression to the mean: reactions do not separate a post that broke out from an ordinary one well. It is strongest on X, where posts above 10 000 views are underestimated about tenfold; out of platform, Instagram underestimates large reels by about ×3.

Intervals. On the same platform the 90% interval covers 87–91% of posts on TikTok, YouTube and X, also when the model is trained on earlier months and tested on a later one (0.90–0.93). Out of platform the spread of the X and YouTube errors alone gives too wide an interval (it covers 96% of Facebook videos). A well-calibrated interval needs calibration on a few brands from the target platform, i.e. a few dozen posts with known views: calibrated on 25% of the brands it covers 89% of Facebook videos, but only 84% for the 70 videos from accounts above one million followers (±9 percentage points). On Instagram only 2 brands are available for calibration, so we do not evaluate it there.

5 Transfer across platforms

Different training sets. We have true views on all five platforms, so each can be treated as "hidden" (Table 3). Facebook and Instagram can be estimated from almost any set with an error of ×2.4–×3.1: on Facebook as well as a ridge trained on it (×2.4), on Instagram ×2.5–×2.8 against ×2.4. Training on X alone is the worst. X is an exception in both directions: a model trained on X is off by ×5.5 on YouTube, and no set reaches an error below ×3.6 on X, against ×3.3 when trained on X. Part of this gap is the early read of views in the export: on re-read data X as the target has ×3.0 instead of ×4.4, and YouTube ×2.2 instead of ×4.1 (medians over sets). Confidence intervals are about ±0.013–0.032 MAE on TikTok, YouTube and X, ±0.04 on Instagram and ±0.07 on Facebook, so differences between sets below about ×1.1–×1.2 on Facebook and Instagram are within noise.

Three ranges instead of a number. This test is entirely out of platform, also for TikTok, YouTube and X, so the model differs from the one in Fig. 2. We cut views into three ranges: below 1 000, 1 000–10 000 and above 10 000 (Fig. 3). Accuracy is 78–81%, but the ranges differ greatly in size, so balanced accuracy (mean recall over ranges) matters more: 78% on TikTok, 76% on Facebook, 70% on YouTube and 65% on Instagram, against 33% for guessing; macro F1 0.70–0.78. We are off by two ranges for 0.2–1.9% of posts. The middle range is usually the weakest (recall 50–73%), on Instagram the top one (49%). On X the model practically fails: the 96% accuracy comes from 95% of posts having fewer than a thousand views, and balanced accuracy is only 46% (of 98 posts above a thousand it gets 23 right). We see two reasons: 62% of X posts have no like (removing them from the training data lowers the error on X by 0.08), and views in the X export are read early, about two times lower than in our re-read.

Table 3: Typical error (×) of the regression on likes and comments with different training platforms.
Ridge on log likes and comments, posts from the training platforms pooled into one set. Row: platforms in training (TT TikTok, YT YouTube, X, FB Facebook, IG Instagram): single ones, X + YouTube (the main set) and all sets of four platforms; the full table of 30 sets is in the repository. Column: test platform, absent from training. First row: the same ridge trained on the test platform itself (brand-grouped CV), not the CatBoost of Fig. 2. Bold: best result in the column among those shown. Darker: larger error.

Trained on	TikTok	YouTube	X	Facebook	Instagram
ridge, same platform	2.0	4.1	3.3	2.4	2.4
YT	**2.0**	train	4.6	2.7	2.6
FB	2.3	4.3	**3.6**	train	**2.5**
IG	2.5	4.5	4.1	2.5	train
TT	train	**4.1**	4.1	2.6	2.6
X	3.4	5.5	train	3.1	2.7
YT + X	2.1	train	train	**2.4**	2.7
YT + X + FB + IG	2.1	train	train	train	train
TT + YT + X + IG	train	train	train	2.5	train
TT + YT + X + FB	train	train	train	train	2.7
TT + X + FB + IG	train	4.1	train	train	train
TT + YT + FB + IG	train	train	4.4	train	train

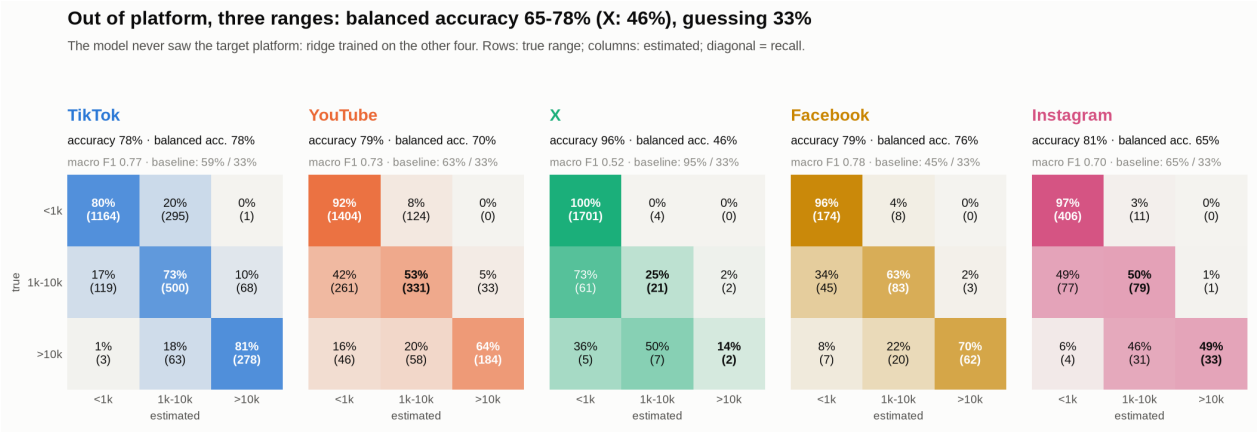


Figure 3: Out-of-platform estimate cut into three view ranges.
For each platform, ridge on likes and comments trained on the other four. Row: true range; column: estimated; in a cell: share of the row and number of posts; the diagonal (bold) is the recall of the range. Above the table: accuracy, balanced accuracy (mean recall), macro F1; grey: values when always guessing the most frequent range.

What the model needs. The transfer works only with likes read at the same moment as the views (Fig. 4A). Comments alone give ×6.3–×7.4, and likes and comments from the export, read at a different moment than the views, ×6.8. On the same platform (Fig. 4B) missing followers hardly hurt, while missing reactions raise the error to ×3.8–×5.2. Therefore we train a separate model with its own interval for each pattern of available fields.

Fresh source data. When the same model is trained on re-read X and YouTube, with reactions from the same moment, the error on Instagram drops from ×2.72 to ×2.41 (by coincidence the same as a ridge trained on Instagram itself), and on Facebook it stays at ×2.44. The underestimation of large posts and of brand totals roughly halves. Merely shifting export views to a later age does not achieve this, so what matters is that views and reactions come from the same moment. This variant does not replace the main result, because the Instagram gain was checked after part of the reels had been seen.

A sixth platform. We fetched through the public API 538 questions from Stack Exchange sites (among others Stack Overflow, Super User, Android) about 48 brands from our data, from April–August 2026. We treat an upvote as a like, and answers and comments as comments; this mapping was fixed before evaluation. The ordering transfers but the level does not: the X and YouTube model is off by ×3.14, worse than a constant median from Stack Exchange itself (×1.96) and than the median of X and YouTube (×2.10), because 96% of questions are underestimated (question views accumulate over months and an upvote is not a like). It keeps the ordering of questions as well as a ridge trained on Stack Exchange (Spearman 0.57 in both cases). Question views have a small spread (standard deviation 0.40 against 1.21 on Facebook), so even a ridge trained on Stack Exchange (×1.77) is only 15% better than the median. The views to reactions ratio thus

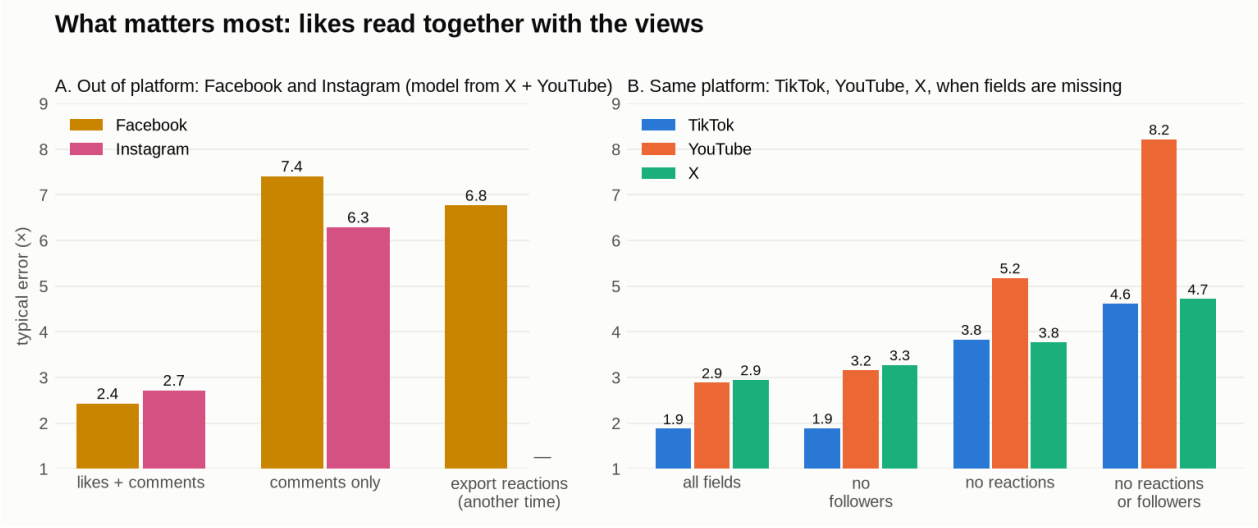


Figure 4: Typical error with different inputs.
A (out of platform): Facebook and Instagram, ridge trained on X and YouTube. "Likes + comments": fields read together with the views; "comments only": a model on comments alone; "export reactions": likes and comments from the Brand24 export, read at a different moment than the views (not available for Instagram). B (same platform): TikTok, YouTube, X, a separate CatBoost for each pattern of available fields, brand-grouped validation.

transfers between video services, and to a question-and-answer site only the ordering does.

6 Comparison of methods

Model families. Table 4 compares models on the same platform, on the same inputs. After tuning, almost all families give the same error. TabPFN v2 [8] beats CatBoost by 0.004–0.009, less than the confidence interval (±0.012–0.021), and takes about 150 times longer. A stack of all families (a second-level model weighting their estimates) improves the result by 0.012–0.029, on YouTube and X more than the noise. We still keep a single model: the gain is ×1.03–×1.07 against a typical error of ×1.9–×3.0, a stack requires maintaining over a dozen models, and the CQR intervals and the transfer are built on one model, while out of platform the linear model wins anyway.

Table 4: Same platform: model families, MAE on the $\log_{10}$ scale and typical error (×).
Followers, reactions and missingness flags, brand-grouped validation, nested tuning (40 settings per fold). TabPFN, TabICL and the stack from separate runs on the same data. 95% CI for CatBoost: ±0.012 / ±0.021 / ±0.021, so differences below about 0.01 are within noise. *: split with no shared brands, authors or texts. The full table of 17 models is in the repository.

Model	TikTok	YouTube	X
platform median views	0.663 (×4.60)	0.944 (×8.79)	0.667 (×4.65)
fixed multiple of followers	0.919 (×8.30)	0.855 (×7.16)	0.879 (×7.57)
fixed likes multiplier*	0.348 (×2.23)	0.609 (×4.06)	0.560 (×3.63)
linear regression (ridge)	0.277 (×1.89)	0.545 (×3.51)	0.487 (×3.07)
random forest	0.275 (×1.88)	0.463 (×2.90)	0.473 (×2.97)
XGBoost [9]	0.273 (×1.87)	0.450 (×2.82)	0.467 (×2.93)
CatBoost (final model)	0.272 (×1.87)	0.452 (×2.83)	0.470 (×2.95)
multilayer perceptron (MLP)	0.276 (×1.89)	0.454 (×2.84)	0.477 (×3.00)
TabPFN v2	0.268 (×1.85)	0.443 (×2.77)	0.465 (×2.92)
TabICL [10]	0.272 (×1.87)	0.445 (×2.79)	0.466 (×2.92)
stack of all families	0.260 (×1.82)	0.425 (×2.66)	0.441 (×2.76)

Extra data and features. We tested more than twenty additional signal sources: text (from TF-IDF to e5 sentence embeddings [11] and the VADER sentiment lexicon [12]), mention type and sentiment labels from a language model, author and brand graphs (GraphSAGE [13], metapath2vec [14]), brand data from Wikipedia and Wikidata, synthetic large posts (SMOTER/SMOGN [15]), representations learned without labels (SCARF [16]) and pseudo-labels. None improves the error by more than ×1.04, most by less than

×1.02, and some make it worse by up to ×1.06. On Facebook text worsens the transfer (MAE 0.385 → 0.436).

Author history. The same author has a similar view surplus (truth minus estimate) on successive posts: the correlation between the first and later posts is 0.68–0.70. Adding the mean surplus of the author's earlier posts to the estimate, shrunk towards zero when there are few posts, lowers the error on those posts by ×1.3–×1.4. This is, however, a different task, because it needs known views of the author on the same platform, and such history exists for only 3–16% of posts (≤0.003 in the mean error). The same author on another platform appears for 0.67% of posts.

7 Applications

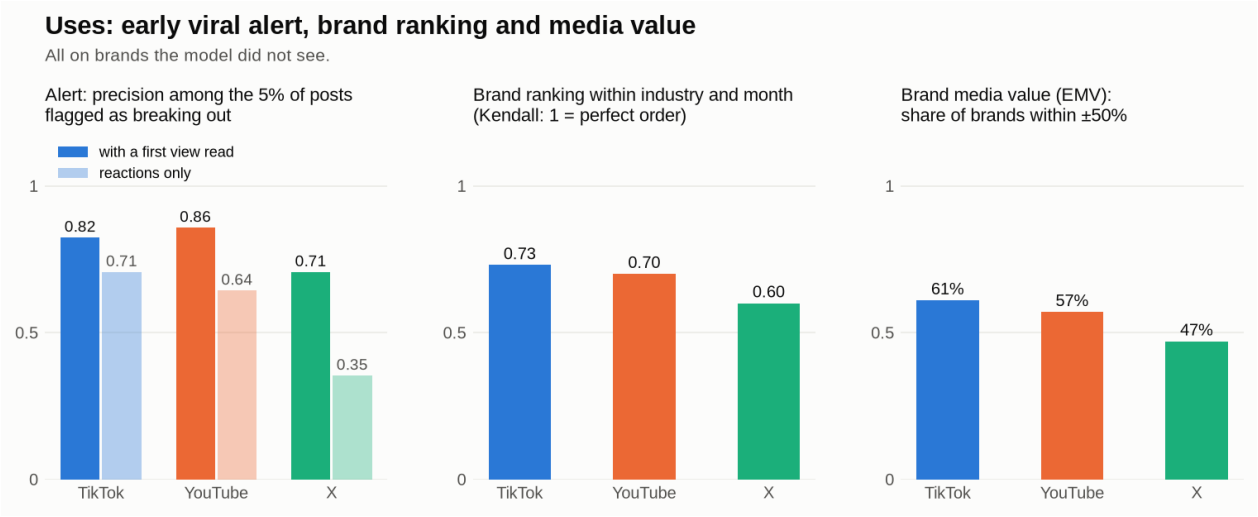


Figure 5: Applications of the estimate on TikTok, YouTube and X (same platform).
Alert: from a first, early read of views and reactions the model flags the 5% of posts that will end in the top 5% of views; the bar is the share of correct flags (289–336 posts per platform, one month). Ranking: mean Kendall rank correlation between the estimated and true ordering of brands in industry–month pairs (52 / 51 / 42 pairs). EMV (earned media value): views converted to money at a fixed rate of 5 USD per thousand views; share of brands whose estimated EMV is within ±50% of the true one (238 / 213 / 189 brands).

Breakout alert. From the first read the model flags the 5% of posts that will end in the top 5% of views, with precision 0.82 / 0.86 / 0.71; the same alert from reactions alone, without the model, gives 0.71 / 0.64 / 0.35 (Fig. 5).

Brand ranking. By summing the estimated views of brands within an industry and month we recover their order (Kendall 0.73 / 0.70 / 0.60).

Media value. A brand's EMV computed from estimated views is within ±50% of the true one for 61% / 57% / 47% of brands. The rate only scales the result, so the shares do not depend on it.

Outlying posts. On TikTok, posts with many more views than their reactions suggest are more often promotion (51% against 34%), and those with many fewer are more often spam (26% against 13%; mention type from language-model labels). **TODO:** fill to a full page: e.g. alert precision versus number of alerts, a ranking example for one industry, mentions not about the brand (E224).

8 Limitations and recommendations

Few brands. The Facebook result rests on 20 brands and the Instagram result on 11; the three largest brands hold 39% and 44% of posts, and without them the error rises to ×2.89 and ×2.96. The error of a single brand ranges from ×1.5 to ×6.2 on Facebook and from ×2.2 to ×4.3 on Instagram. With so few brands the brand-bootstrap confidence intervals may be too narrow.

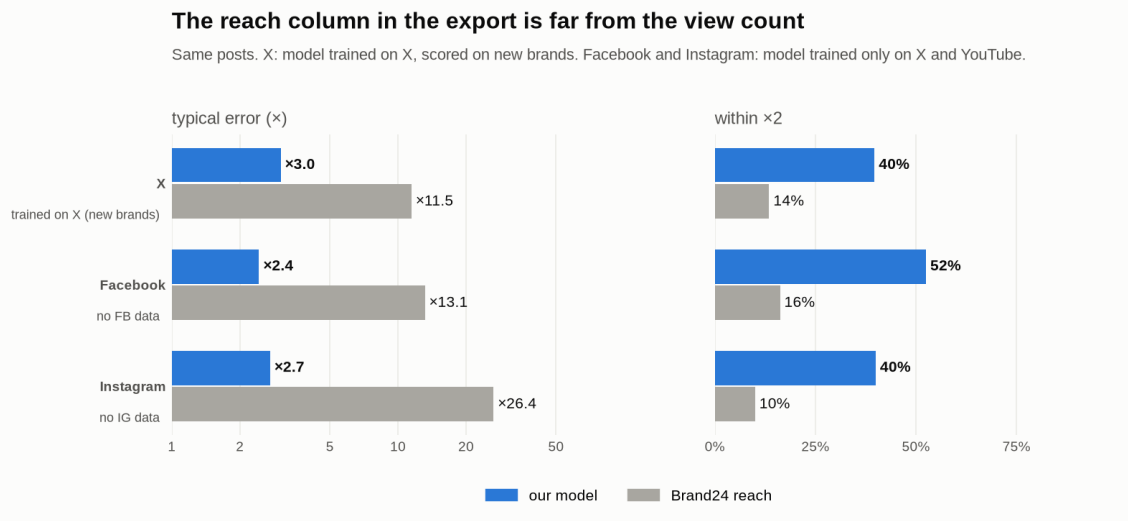


Figure 6: Our estimate versus the reach column in the Brand24 export, on the same posts.
Only platforms where reach is not the view count. Reach is a different measure from views, so the figure shows how far from the views the number a client gets today is, not a comparison of two models. X (1 803 posts): model trained on X, evaluated on brands and authors it has not seen; truth from the export. Facebook (402 videos) and Instagram (642 reels): out-of-platform model trained only on X and YouTube; truth from our read.

Post selection. We measure only videos and reels that were still available. On Instagram, reels with hidden likes were also dropped, exactly the ones where an estimate would be most needed in practice; there the model would have to rely on comments alone (×6.3).

Read age. Views in the X and YouTube export were read at a median of 23 days after publication, our Facebook and Instagram reads after 310 and 334 days. The agreement on Facebook may thus partly come from two effects cancelling: moving only the source views to a later age raises the Facebook error by 0.10 (×3.05), and only moving views together with reactions restores ×2.44.

Mentions not about the brand. Brand24 matches mentions by keywords. In 6.7% of mentions the brand name does not appear in the text at all, and according to separate topic labels from a language model (not yet checked by a person) the brand is not the main topic in about 64% of mentions. This does not change the error of per-post view estimates, but brand totals also include posts about something else, and the brand-grouped validation split is less clean.

Other. On the same platform the most popular 1% of posts on YouTube and X is underestimated ×9–×14, so brand totals are off by more than ×2 for 29–47% of brands. Almost every post has a different author (2 411 authors for 2 491 posts on TikTok, 2 041 for 2 441 on YouTube, 1 658 for 1 803 on X), so author history cannot be used for most posts. The results describe associations, not causes. Code, numbers and cards of all experiments are in the project repository.

Recommendations. We recommend to the monitoring provider: read reactions together with views, record missing values as missing rather than zero, take a second read of views, and keep author histories. Fig. 6 shows what this gives today: the reach column in the export, read by clients as the number of people reached, is further from the views than our estimate.

Use of AI and reproducibility. **TODO:** to be written by the team in their own words. Facts to include: AI agents (Claude Code, DeepSeek) wrote the analysis and data-reading code, ran the experiments, made the figures and a draft of the text; language models labelled mention type and sentiment; the agents' session logs are in `logs/`. Every result has an experiment card, a row in the register (`docs/en/experiments.md`), a script in `src/` and numbers in `data/processed/`.

References

[1] A. Tatar, M. D. de Amorim, S. Fdida, P. Antoniadis. A survey on predicting the popularity of web content. *Journal of Internet Services and Applications*, 5:8, 2014.

[2] B. Shulman, A. Sharma, D. Cosley. Predictability of popularity: gaps between prediction and understanding. *ICWSM*, 2016.

[3] S. D. Roy, T. Mei, W. Zeng, S. Li. Towards cross-domain learning for social video popularity prediction. *IEEE Transactions on Multimedia*, 15(6), 2013.

[4] S. Ben-David, J. Blitzer, K. Crammer, A. Kulesza, F. Pereira, J. W. Vaughan. A theory of learning from different domains. *Machine Learning*, 79, 2010.

[5] L. Prokhorenkova, G. Gusev, A. Vorobev, A. V. Dorogush, A. Gulin. CatBoost: unbiased boosting with categorical features. *NeurIPS*, 2018.

[6] A. E. Hoerl, R. W. Kennard. Ridge regression: biased estimation for nonorthogonal problems. *Technometrics*, 12(1), 1970.

[7] Y. Romano, E. Patterson, E. Candès. Conformalized quantile regression. *NeurIPS*, 2019.

[8] N. Hollmann et al. Accurate predictions on small data with a tabular foundation model. *Nature*, 637, 2025.

[9] T. Chen, C. Guestrin. XGBoost: a scalable tree boosting system. *KDD*, 2016.

[10] J. Qu, D. Holzmüller, G. Varoquaux, M. Le Morvan. TabICL: a tabular foundation model for in-context learning on large data. *ICML*, 2025.

[11] L. Wang et al. Text embeddings by weakly-supervised contrastive pre-training. arXiv:2212.03533, 2022.

[12] C. J. Hutto, E. Gilbert. VADER: a parsimonious rule-based model for sentiment analysis of social media text. *ICWSM*, 2014.

[13] W. L. Hamilton, R. Ying, J. Leskovec. Inductive representation learning on large graphs. *NeurIPS*, 2017.

[14] Y. Dong, N. V. Chawla, A. Swami. metapath2vec: scalable representation learning for heterogeneous networks. *KDD*, 2017.

[15] P. Branco, L. Torgo, R. P. Ribeiro. SMOGN: a pre-processing approach for imbalanced regression. *LIDTA*, PMLR 74, 2017.

[16] D. Bahri, H. Jiang, Y. Tay, D. Metzler. SCARF: self-supervised contrastive learning using random feature corruption. *ICLR*, 2022.